

CEGAL

Robust Patching on EXACC & EXACS , ZDLRA backup solution

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More than 15 years of experience in various domains such as Banking, Energy, Finance, Shipping, Automobiles and Retail where working with managing infrastructure, security, cloud and cloud migration project.

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More than 27 year's experience with Oracle Database and Middleware, and recently involved in Oracle Cloud projects.

Alumni's: Ace Associate and OUGN board

Content

Introduction

Method

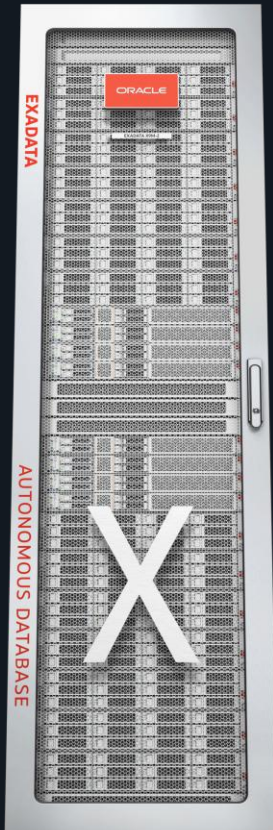
Patching

Business case

Challenges

Solution

Introduction



Ideal Database Hardware

Scale-out, database optimized compute, networking, and storage

Database Aware System Software

Unique algorithms vastly improve OLTP, Analytics, and Consolidation

Automated Management

Fully automated and optimized end-to-end

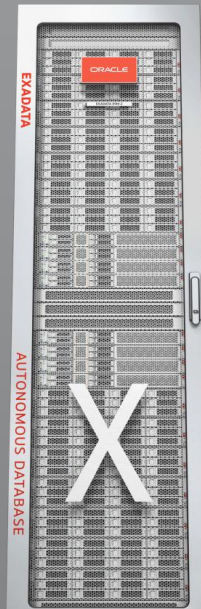
Available

On Premises

Cloud@Customer

Oracle Cloud

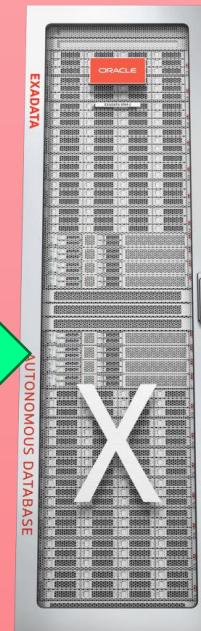
PRODUCTION



Database Database Database
130+ databases

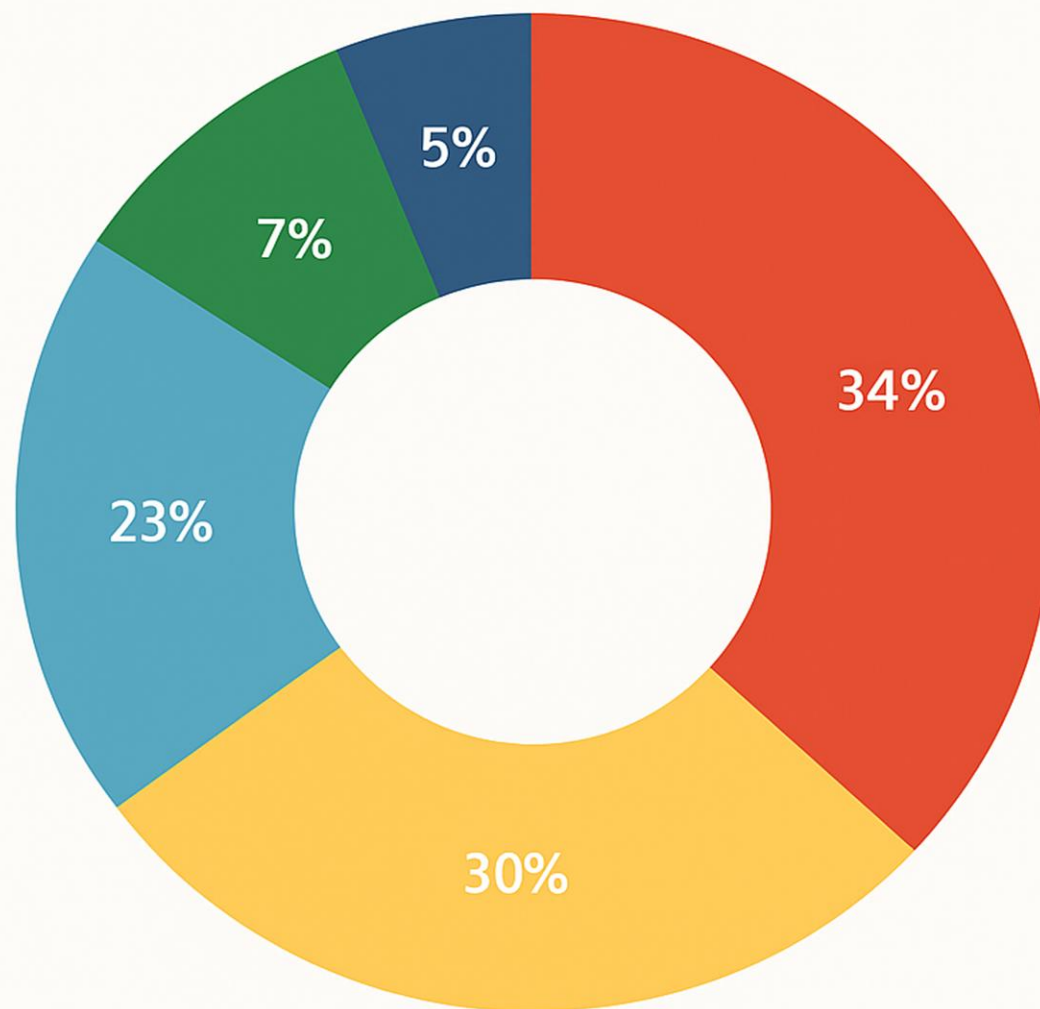
ASYNC

TEST/STANDBY



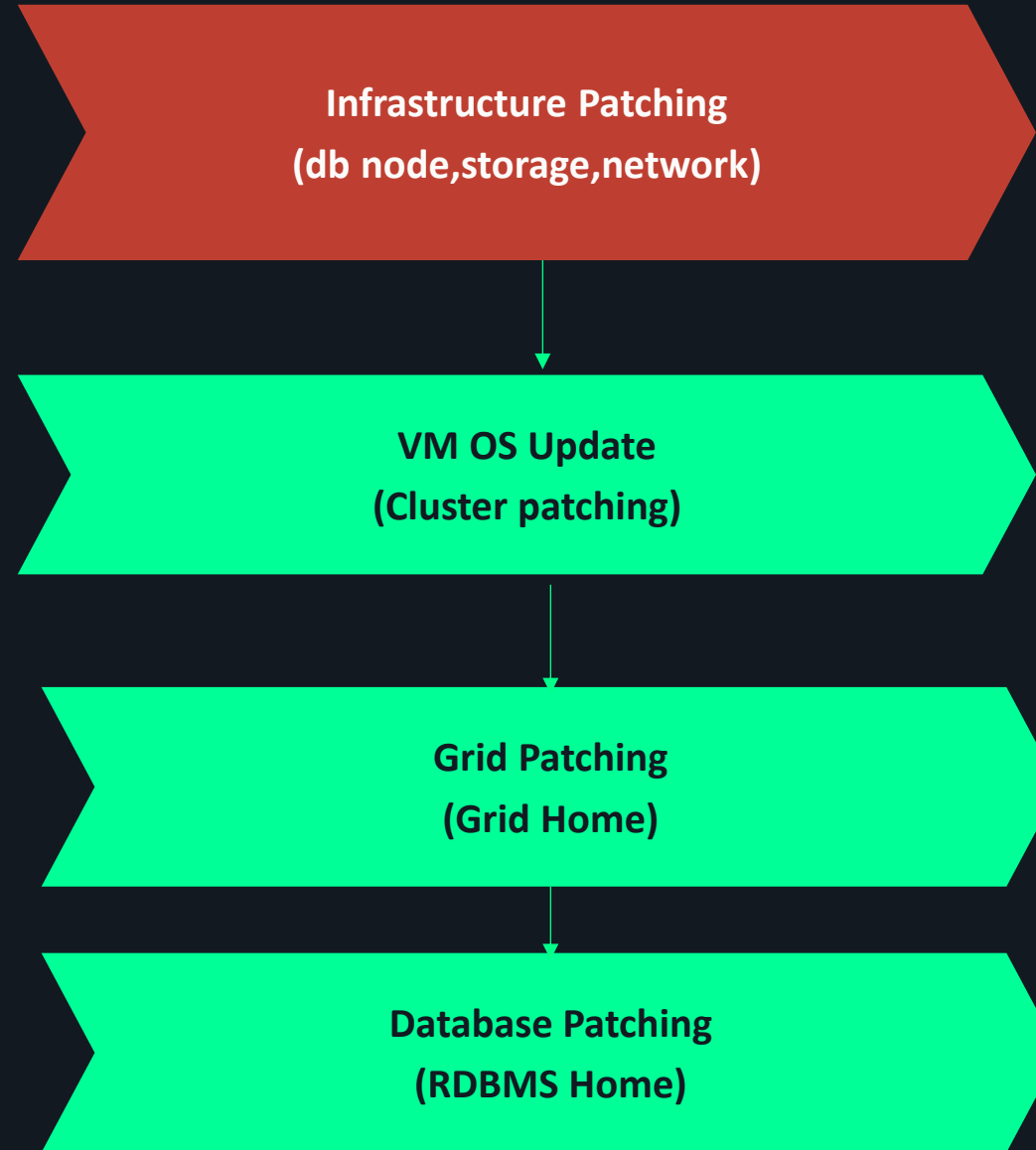
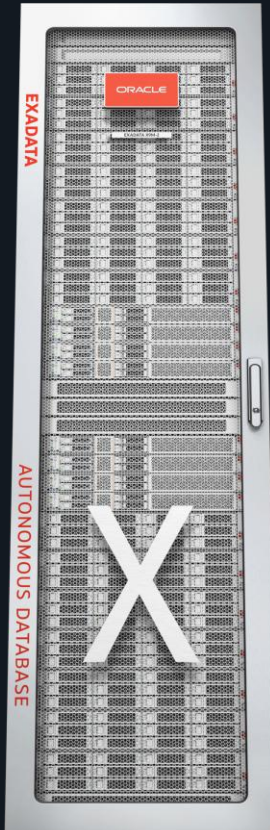
Top Pain Points

- Upgrades
- Patching
- High Availability
- Cloud Migration
- Backup and Recovery



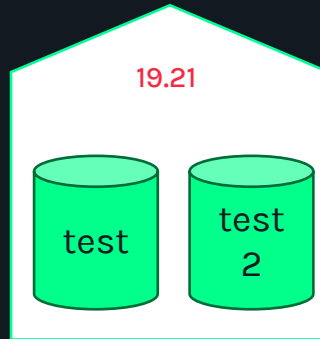
MAA can solve pain points

Types of Patching

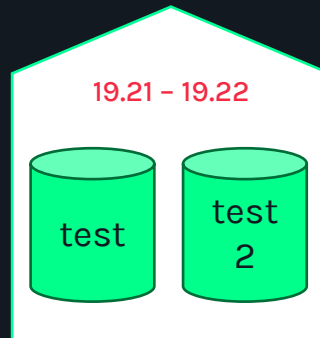


Method (In Place Patching)

1) Multiple Database Running from an Oracle Home



2) Patching all the databases in Oracle Home(Downtime)



3) Patched database up and running in the Oracle Home And Apply Sqls to the instances

Advantages

- No Extra Disk Space Requirement
- Less Configuration Work
- Very Comfortable

Disadvantages

- Higher Risk
- Difficult Rollback
- Longer Downtime
- No Parallel Testing
- Limited Flexibility

Method (Out of Place Patching)



Advantages

- Minimal Downtime
- Safe Rollback
- Testing Before Cutover
- Reduced Risk

Disadvantages

- More Disk Space Needed
- Manual Configuration Steps



```
graph TD; C((Challenges)) --- MD[Multiple Databases]; C --- D[Downtime]; C --- TC[Time Consuming]; C --- PQ[Patching Quarterly(Databases and Grid)];
```

Challenges

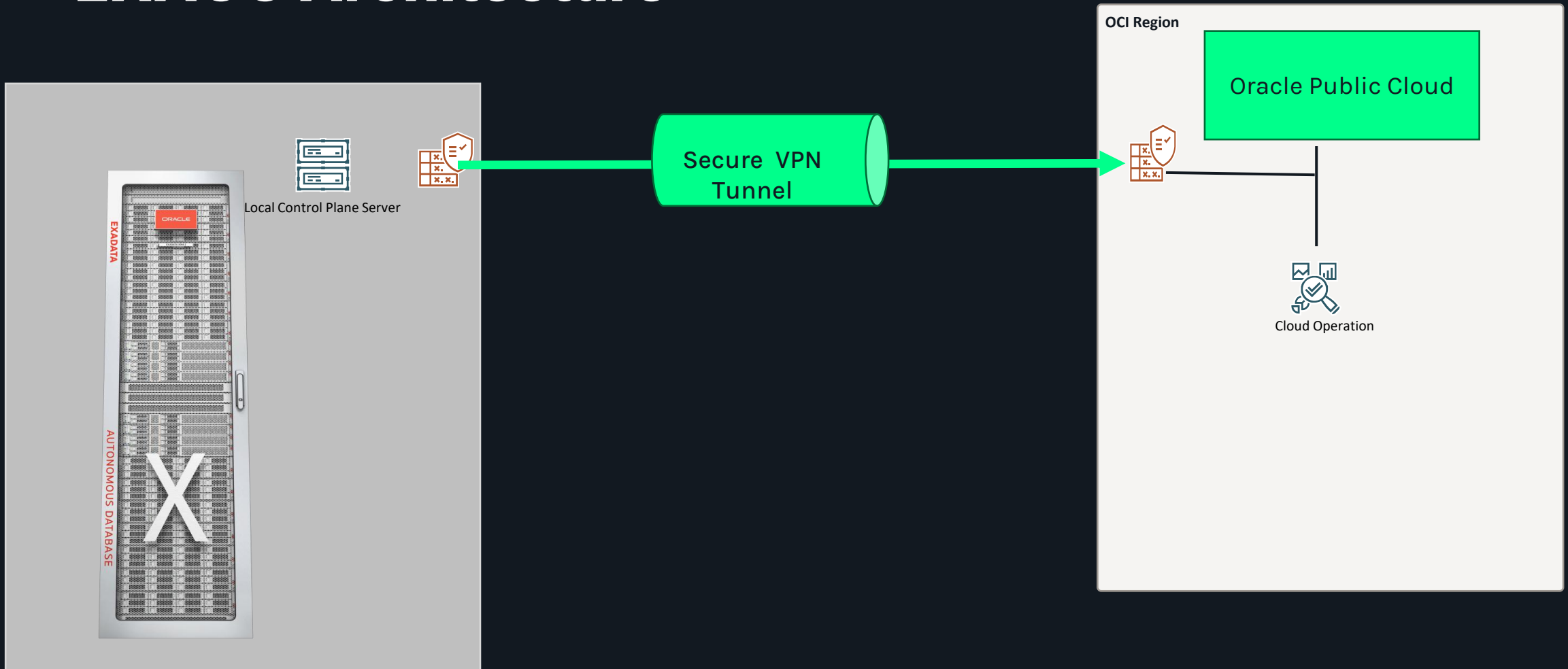
Multiple Databases

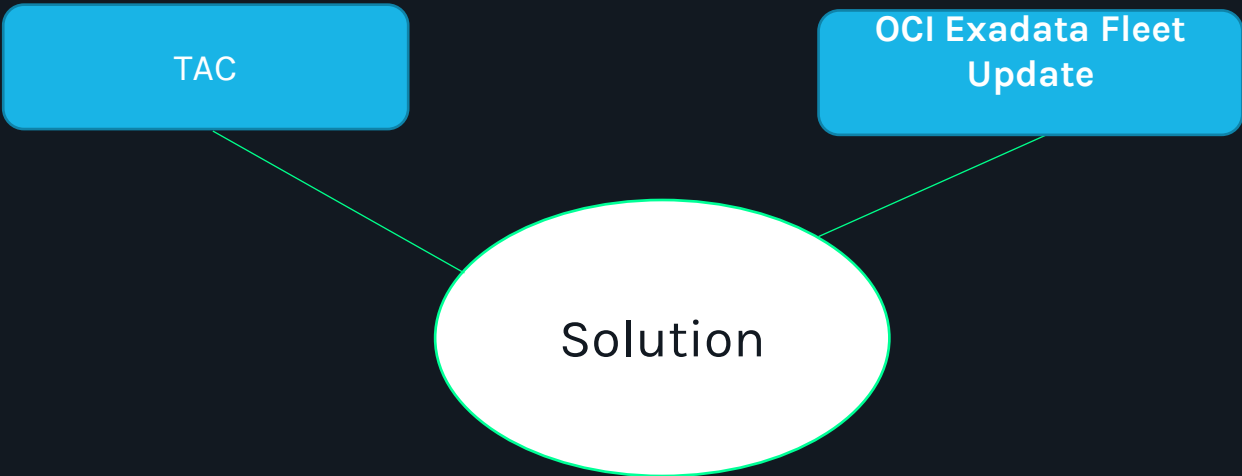
Downtime

Time Consuming

Patching Quarterly(Databases
and Grid)

EXACC Architecture

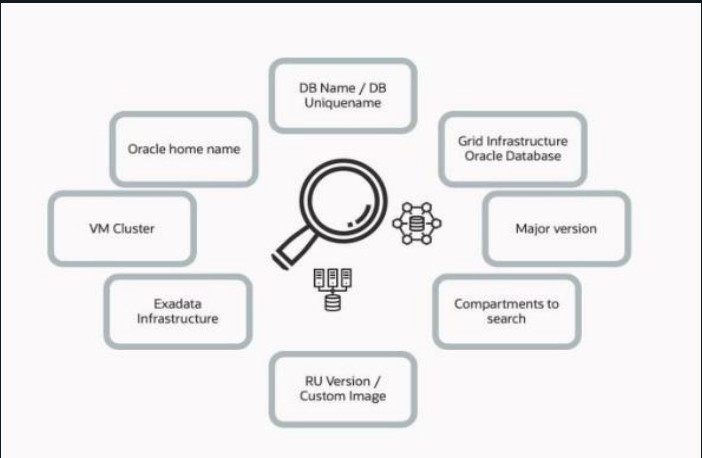




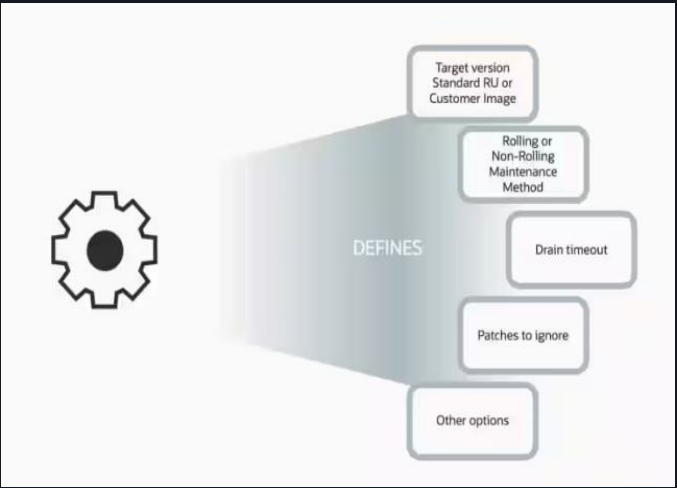
Pre-requisite:

- IAM policy(to access vm,db and infra)
- Create database home .

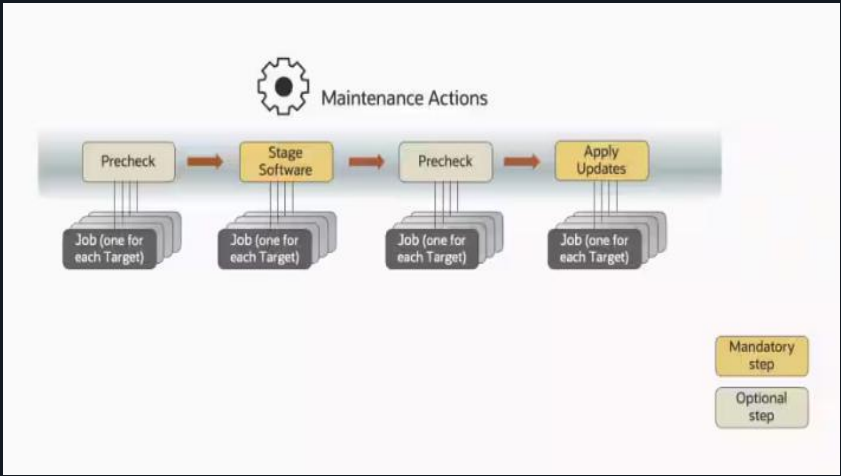
1) Create Collection



2) Create Maintainance Cycle



3) Maintainance Execution



Exadata Fleet

2025-05-09 10:59 UTC

Recorded by

Jon Petter Hjulstad

Organized by

Jon Petter Hjulstad

Benefits



Significant time savings by patching multiple databases simultaneously



Once scheduling is set up, only light monitoring is required



Rolling patching minimizes downtime for apps not MAA-configured

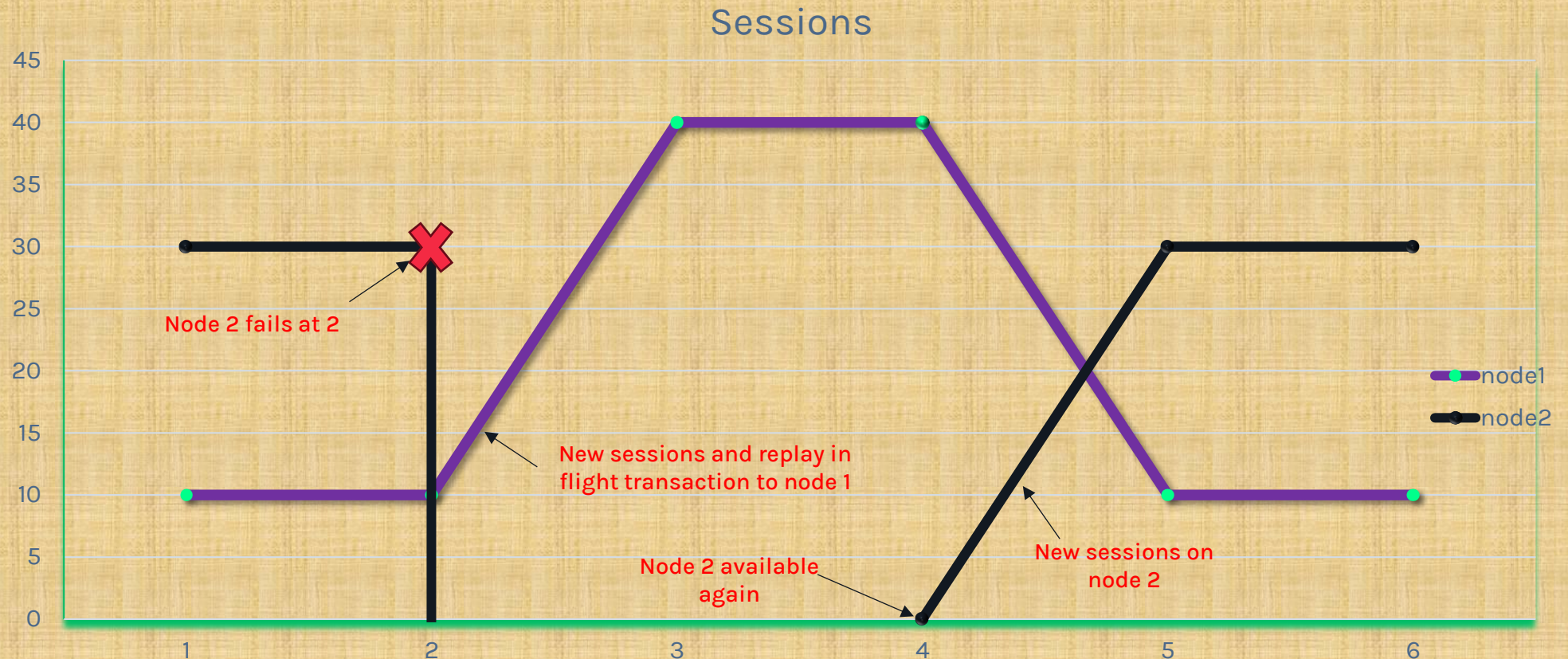


Close collaboration with Oracle (Product Owner & Dev team) enhanced results

Time Savings and Impact

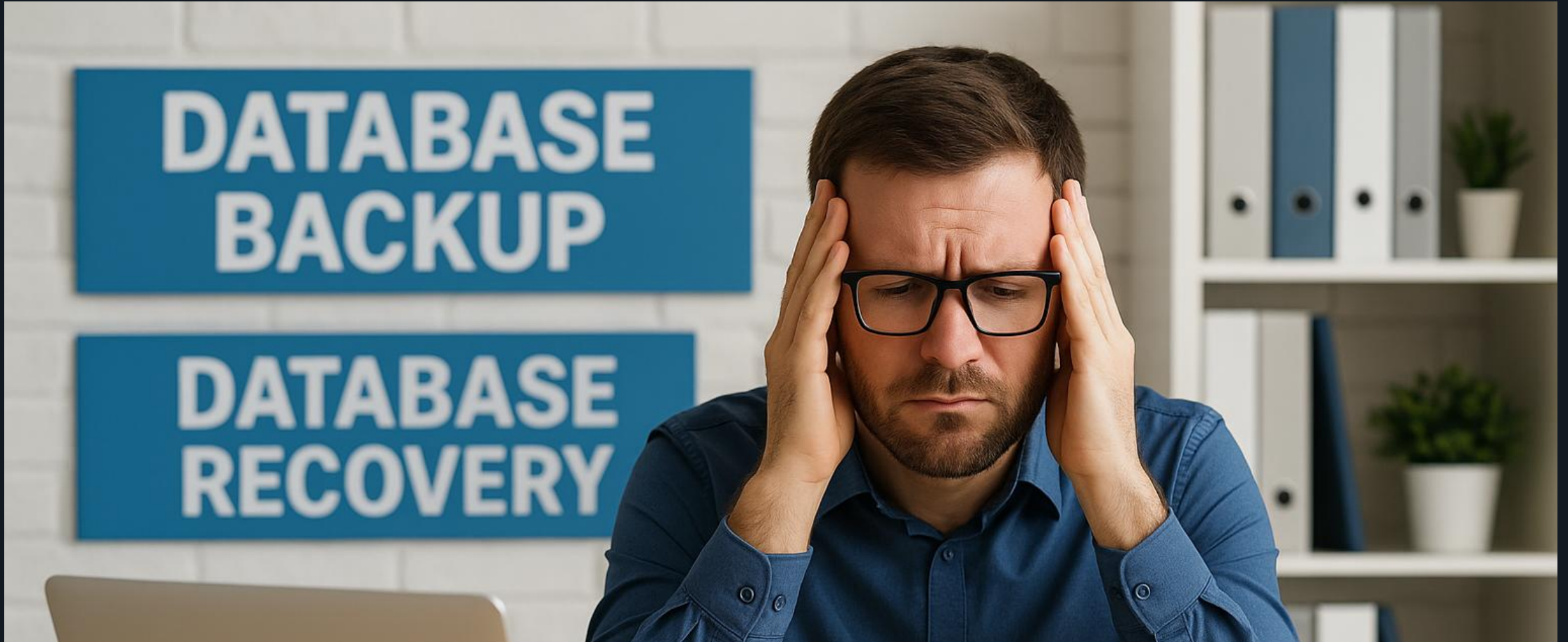
- Patching 1 DB manually: ~1.5 hours
- Last patch run: 8 DBs in ~1.5 hours (in parallel)
- Runtime alone saved: ~10.5 hours
- Additional time saved: fewer checks, coordination, and manual steps
- Organizational shift toward more automated and online operations

Transparent Application Continuity



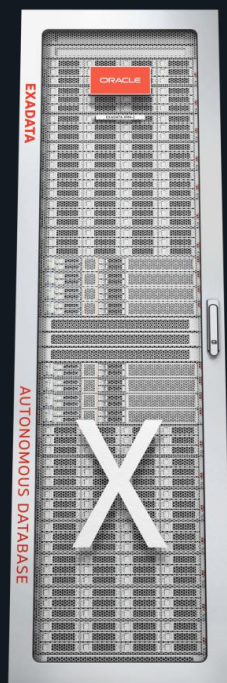
```
srvctl add service -db test -service test_tac_ha -preferred test1,test2 -failover_restore AUTO -failoverretry 30 -failoverdelay 10 -commit_outcome TRUE -failovertype AUTO -replay_init_time 1800 -retention 86400 -drain_timeout 300 -stopoption IMMEDIATE
```

Backup and Recovery



ZDLRA (Zero Data Loss Recovery Appliance)

Async redo transport(no archivelog backup job needed)
Based on Dataguard Technology
Validated and stored as Archivelog Backup
Very Low impact on Production
Ability to fetch Archive log after a Connection loss



Redo Blocks

RFS

Rman Backups

HTTP Servlet

RMAN Backup (via HTTP)
Backups are written to Flash
Backups are Indexed
Compressed and written as Virtual Fulls
Incremental Forever Strategy



Target Name	Database Size (GB)	Recovery Window			Unprotected Data Window		Errors/Warnings	Redo Shipping	Copy-to-Tape	Replication	Last Complete Backup
		Goal (days)	Current (days)	Needed Space (GB)	Threshold	Current					
store32	573.0	2.0	6.2	465.4	24.0 hrs	5.7 hrs		✓	✓		Sep 4, 11:12 PM PDT
store33	473.0	2.0	6.4	449.0	24.0 hrs	< 1 sec		✓	✓		Sep 4, 11:16 PM PDT
store34	463.0	2.0	6.4	444.5	24.0 hrs	< 1 sec		✓	✓		Sep 4, 11:19 PM PDT

Feature / Aspect	General Purpose Backup Appliance	Oracle Recovery Appliance (ZDLRA)
Purpose	Generic backup solution for various data types and platforms	Purpose-built appliance for Oracle database backups and recovery
Supported Data Types	Supports file systems, VMs, databases (Oracle, SQL Server, etc.)	Exclusively optimized for Oracle databases
Backup Method	Typically uses RMAN or third-party software	Deep integration with RMAN; supports incremental forever strategy
Data Loss Protection	Possible data loss between last backup and failure	Zero data loss capability with Real-Time Redo transport
Real-Time Redo Capture	Not supported	Supported – captures and stores redo in real time
Deduplication & Compression	Software- or hardware-based, general deduplication	Oracle-aware deduplication
Restore Speed & Efficiency	General performance; varies by configuration	Optimized for fast Oracle database restores
Validation of Backups	Manual or periodic validation	Automatic validation of backups via RMAN
Scalability	Scalable, but with generic architecture	Scales linearly with Oracle database workload
Security Features	General encryption and access control	End-to-end data protection; tightly integrated with Oracle security stack
Automation & Policy Management	Generic automation, often third-party	Built-in automation for backup, retention, validation, and replication
Integration with Oracle Ecosystem	Limited or via plugins	Full native integration with Oracle Enterprise Manager, RAC, Data Guard
Use Case	Mixed IT environments with varied data types	Enterprise-grade Oracle-only data centers needing zero data loss backup

Thank you

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